A
Case Study Report on :
"grocery shop billing system"

PROJECT SYNOPSIS

PROBLEM STATEMENT:

The question is about to create a billing system for the shopkeepers so that they can create a bill easily and quickly. The system should be simple to give comfort to both the shopkeepers and customers while understand the bill.

Aim:

To make software fast in processing, with good user interface so that user can change it and it should be used for a long time without error and maintenance

Objective:

- *To reduce the time for organisation.*
- *To increase efficiency and accuracy of the system.*
- *To reduce pressure on the labour and relieving man power from repetitive and dull job.*
- *To make the retrieval of information faster.*
- *To make the system more feasible*
- *To reduce large amount of paper work.*
- *To make the system more reliable and to avoid any ambiguity.*

- *To reduce the cost factor of the system.*
- *To make the system more flexible.*
- *It provides a convenient solution of billing pattern.*
- *It maintains new entered category or product*

Introduction

Domain:

Grocery shop billing system is the system to automate the process of ordering and billing of a grocery store. Grocery is the place where customers come to purchase their daily using products and pay for that. So there is a need to calculate how many products are sold and to generate the bill for the customer.

This system is built for fast data processing and bill generation for customers. It also allows the Grocery customer to purchase and pay for the items purchased. The users will consume less time in calculation and the sales activity will be completed within a fraction of

seconds whereas in a manual system will make the user to write it down which is a long procedure and it also consumes a lot of time. Because of this software, paper work will be reduced and the user can spend more time on monitoring the Grocery.

Conclusion:

The main purpose of online booking of railway tickets is to provide a convenient and hasslefree booking experience for customers. Online booking allows customers to book train tickets from anywhere at any time using a computer or mobile device with an internet connection. This eliminates the need for customers to visit railway stations or booking counters, saving them time and effort

Requirement specification

Software requirements:

Turbo : c language

Hardware requirements:

Processor: intel core i5

Ram:16gb ,

REFRENCES:GOOGLE,CHAT GPT

Future Enhancements:

- **Artificial Intelligence (AI) and Machine Learning (ML)** - The use of AI and ML can enhance the user experience by providing personalized and improving the accuracy of bill.

- **Mobile Application** - The development of a mobile application can provide a more convenient and user-friendly platform for shopkeepers to use it when and where ever possible

.

- **Digital Payments** - The adoption of digital payment methods such as e-wallets, UPI, and cryptocurrencies can provide a more secure and efficient payment option for customers.

Source-Code: -

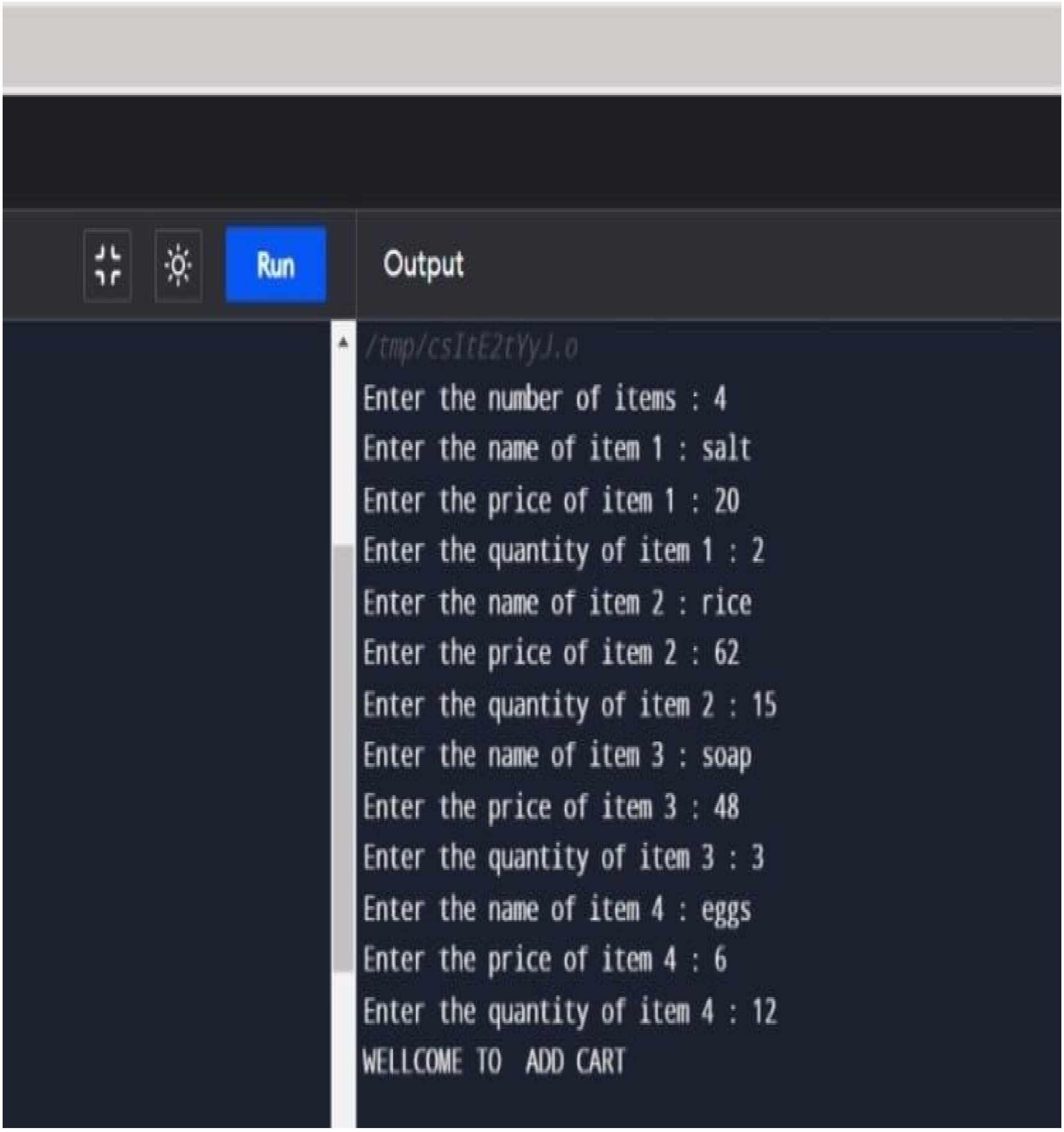
```
#include <stdio.h>
#include <stdlib.h>
struct Item
 {
   char name[50];
   float price;
   int quantity;
};

float calculateTotal(struct Item items[], int
itemCount)
{
   float total = 0;
   for (int i = 0; i<itemCount; i++)
      {
      total += items[i].price * items[i].quantity;
      }
```

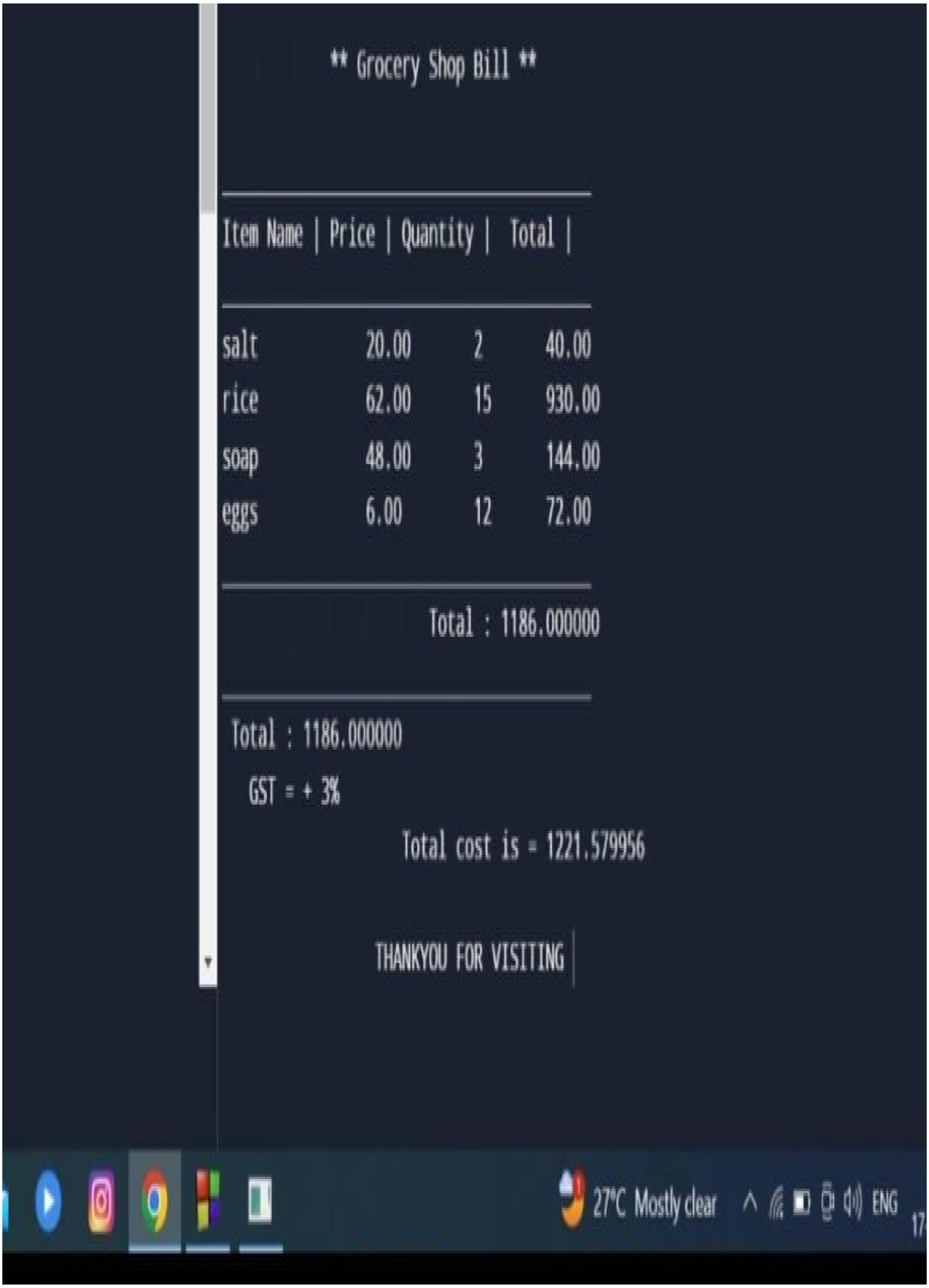
```
    return total;
}
int main()
  {
    int itemCount;
float temp=0;
printf("Enter the number of items : ");
scanf("%d", &itemCount);
    if (itemCount<= 0)
        {
printf("Invalid number of items.\n");
        return 1;
        }

    struct Item items[itemCount];

    for (int i = 0; i<itemCount; i++)
        {
printf("Enter the name of item %d : ", i + 1);
```

```
scanf("%s", items[i].name);
printf("Enter the price of item %d : ", i + 1);
scanf("%f", &items[i].price);
printf("Enter the quantity of item %d : ", i + 1);
scanf("%d", &items[i].quantity);
        }

printf("\n\t\t WELLCOME TO  ADD CART\n");

printf("\n\t\t\t** Grocery Shop Bill **\n");

printf("\n_____________________________________________
_________\n");
printf("Item Name |\tPrice |\tQuantity |\tTotal
|\n");
printf("_______________________________________________
_______\n");
    for (int i = 0; i<itemCount; i++)
    {
```

```
printf("%s\t\t\t%.2f\t\t%d\t\t%.2f\n",
items[i].name, items[i].price, items[i].quantity,
items[i].price * items[i].quantity);
    }
printf("_______________________________________________
_______\n");
temp = calculateTotal(items, itemCount);
printf("                        Total : %f\n",temp);
printf("_______________________________________________
_______");
float gst=(temp/100)*3;
gst = gst+temp;

printf("\n Total : %f\n",temp);
printf("   GST = + 3% \n");
printf(" \t \t\t\t\tTotal cost is = %f",gst);
printf("\n\n    \t\t     THANKYOU FOR VISITING ");

    return 0;
}
```

Output 1:


Run
Output
/tmp/csItE2tYyJ.o
Enter the number of items : 4
Enter the name of item 1 : salt
Enter the price of item 1 : 20
Enter the quantity of item 1 : 2
Enter the name of item 2 : rice
Enter the price of item 2 : 62
Enter the quantity of item 2 : 15
Enter the name of item 3 : soap
Enter the price of item 3 : 48
Enter the quantity of item 3 : 3
Enter the name of item 4 : eggs
Enter the price of item 4 : 6
Enter the quantity of item 4 : 12
WELLCOME TO ADD CART


** Grocery Shop Bill **
Item Name | Price | Quantity | Total |
salt 20.00 2 40.00
rice 62.00 15 930.00
soap 48.00 3 144.00
eggs 6.00 12 72.00
Total : 1186.000000
Total : 1186.000000
GST = + 3%
Total cost is = 1221.579956
THANKYOU FOR VISITING
27°C Mostly clear
ENG

Output 2:


Run
Output
/tmp/csItE2tYyJ.o
Enter the number of items : e
Invalid number of items.

PROGRAM EXPLANATION

1)Include necessary libraries :

#include <stdio.h>

#include <stdlib.h>

This code includes two standard C libraries, stdio.h for input and output functions and stdlib.h for dynamic memory allocation.

2)Define a structure for storing item information:

```
struct Item
{
   char name[50];
   float price;
   int quantity;
};
```

This structure is used to store information about each grocery item, including its name, price, and quantity.

3)Create a function to calculate the total cost of an item:

```
float calculateItemTotal(struct Item item)
 {
   return item.price * item.quantity;
}
```

This function takes an Item structure as an argument and returns the total cost of that item (price * quantity).

4) The main function:

```
int main()
```

This is the main function of the program where execution starts.

5) Declare variables and allocate memory:

```
int numItems;
float totalCost = 0;
```

numItems stores the number of items the user will input, and totalCost keeps track of the overall bill total.

printf("Enter the number of items: ");

scanf("%d", &numItems);

This prompts the user to enter the number of items and reads that value into the numItems variable.

struct Item *items = (struct Item *)malloc(numItems * sizeof(struct Item));

This dynamically allocates memory for an array of Item structures to store the details of the grocery items.

6) Input item details in a loop:

for (int i = 0; i < numItems; i++)

{

// Prompt the user for item details (name, price, and quantity)

// Calculate the total cost for the item and add it to the overall totalCost

}

The program enters a loop to input the details of each grocery item, including its name, price, and quantity. It then calculates the total cost for each item and adds it to the totalCost variable.

7) Print the bill:

```
printf("\nBill:\n");
printf("----------------------------\n");
printf("Item Name   Price   Quantity  Total\n");
printf("----------------------------\n");
for (int i = 0; i < numItems; i++) {
   // Print each item's details (name, price, quantity, and total cost)
}
printf("----------------------------\n");
```

```
printf("Total: $%.2f\n", totalCost);
```

After entering all item details, the program prints a formatted bill that includes each item's name, price, quantity, and total cost. It also displays the overall total cost at the end.

Running Procedure:-

- Install Turbo-C++ in the system and set the path.
- Create a c file with the given name.
- Write the given code in the file created
- Compile it and then run it.

Thank you